

experiment-10

November 11, 2025

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[1]: # --- Step 1: Import Required Libraries ---  
# (from exp 10.pdf, Cell 1)  
import pandas as pd  
import numpy as np  
from sklearn.datasets import load_iris  
from sklearn.model_selection import train_test_split  
from sklearn.preprocessing import StandardScaler  
from sklearn.linear_model import Perceptron  
from sklearn.metrics import accuracy_score, classification_report,  
    ↪confusion_matrix  
import seaborn as sns  
import matplotlib.pyplot as plt  
  
# --- Step 2: Load Dataset ---  
# (from exp 10.pdf, Cell 2)  
iris = load_iris()  
X = iris.data  
y = iris.target  
  
# --- Step 3: Simplify to a Binary Classification Problem ---  
# (from exp 10.pdf, Cell 3)  
# Remove class 2 (Virginica) to make it a binary problem (Setosa vs. Versicolor)  
X_simple = X[y != 2]  
y_simple = y[y != 2]  
  
# --- Step 4: Split Data into Training and Testing Sets ---  
# (from exp 10.pdf, Cell 4)  
X_train, X_test, y_train, y_test = train_test_split(  
    X_simple, y_simple, test_size=0.3, random_state=42, stratify=y_simple  
)  
  
# --- Step 5 & 6: Feature Scaling ---  
# (from exp 10.pdf, Cell 5 & 6)  
scaler = StandardScaler()  
X_train_scaled = scaler.fit_transform(X_train)  
X_test_scaled = scaler.transform(X_test)
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# --- Step 7: Define and Train the Perceptron Model ---
# (from exp 10.pdf, Cell 7)
perceptron = Perceptron(max_iter=1000, eta0=0.1, random_state=42)
perceptron.fit(X_train_scaled, y_train)

# --- Step 8: Make Predictions and Evaluate the Model ---
# (from exp 10.pdf, Cell 8)
y_pred = perceptron.predict(X_test_scaled)

accuracy = accuracy_score(y_test, y_pred)
print(f"\nModel Accuracy: {accuracy * 100:.2f}%")

print("\n--- Classification Report ---")
# Use the target names for the two classes we kept (0 and 1)
print(classification_report(y_test, y_pred, target_names=['Setosa', 'Versicolor']))

print("\n--- Confusion Matrix ---")
cm = confusion_matrix(y_test, y_pred)
print(cm)

# --- Step 9: Visualize the Confusion Matrix ---
# (from exp 10.pdf, Cell 8)
plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Setosa', 'Versicolor'],
            yticklabels=['Setosa', 'Versicolor'])
plt.title('Confusion Matrix')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.show()

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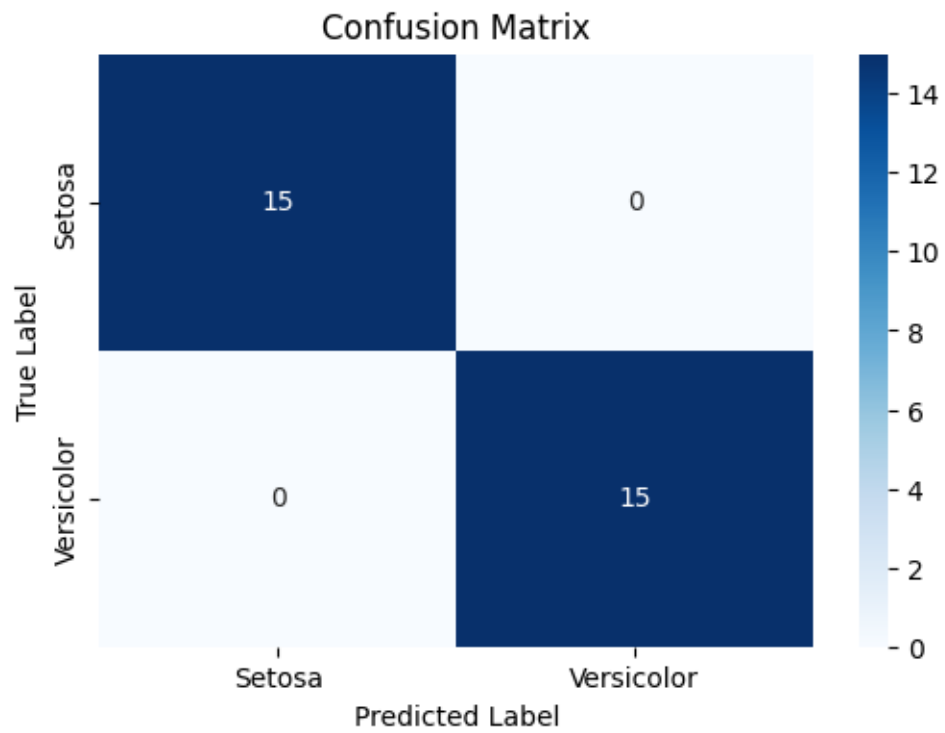
Model Accuracy: 100.00%

--- Classification Report ---

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	15
Versicolor	1.00	1.00	1.00	15
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

--- Confusion Matrix ---

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[[15  0]
 [ 0 15]]
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